

IPC144 – Introduction to C Programming

Practice Exercises

Week 1

Q1 - SWAPPING THE CONTENTS OF TWO VARIABLES

Swapping of two variables means exchanging the values of two variables. For example, let a and b be two given variables with the values 5 and 10 stored in them respectively, as shown in the figures, below.



Before Swapping

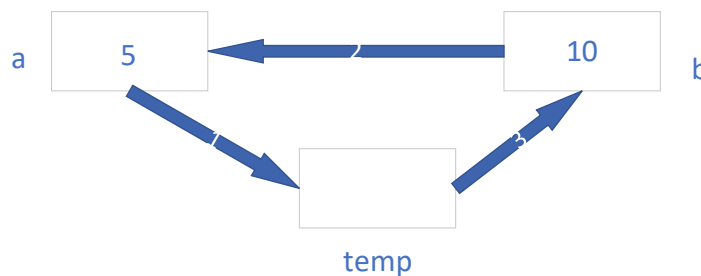
After swapping the value of a will be 10 and the value of b will be 5, as shown in the figures, below



After Swapping

The Method

In order to exchange the values of a and b, first a copy of A is made and stored in a temporary location (temp). Then the value of b is copied on to A. Finally, the contents of temp is copied on to the variable b. These operations are illustrated in the following diagram where 1, 2 and 3 indicate the order in which the operations are performed.



Based on what's given, prepare flowchart and develop pseudo code.

Q2 - SIMPLE INTEREST CALCULATION

The Method

The formula used to calculate simple interest is **$si = (p \times r \times t) / 100$**

where p is the principal,

r is the rate of interest and

t is the time period.

Obviously, the values of p, r and t are input and the total interest for the given period is computed and output. The process involved is shown in the adjoining flowchart.

Care should be taken to avoid integer division while evaluating expressions like

$$si = (p * t * r) / 100.$$

Based on what's given, prepare flowchart and develop pseudo code.

Q3 - COMPUTING THE AREA OF A CIRCLE GIVEN ITS RADIUS

The Method

The relation used is **$area = \pi r^2$** where r is the radius of the circle. It may be observed that here the value of the radius 'r' is to be input. The value of π is 3.14. The process involved can be shown pictorially as in the following flowchart. It should be observed that π is a constant and cannot be used as it is. Its value has been used at the proper place.

Based on what's given, prepare flowchart and develop pseudo code.

Q4 - DETERMINING WHETHER A GIVEN INTEGER NUMBER IS POSITIVE OR NEGATIVE

The Method

We know that a given number could be positive, negative or zero. Though some treat zero as positive, I have preferred it to be zero itself, so that we can demonstrate the use of nested ifs. The procedure here is to first verify whether a given number is zero or not. If not, the procedure verifies whether the given number is less than zero or greater than zero. If less than zero, then the number is negative. If greater than zero, then the number is positive.

Based on what's given, prepare flowchart and develop pseudo code.

Q5 - DETERMINING WHETHER A GIVEN INTEGER NUMBER IS ODD OR EVEN

The Method

The technique used to find out whether a given integer number is odd or even is to modulo divide the given integer by 2 and to verify the remainder. If the remainder is zero then the number is even, else the number is odd.

Based on what's given, prepare flowchart and develop pseudo code.

Q6 - LARGEST OF GIVEN THREE NUMBERS

The Method

Let a, b and c are the given three integer numbers. First read in all the three numbers. Compare a with b to find out whether it is greater than (or equal to) b. If it is true then compare a with c to find out whether it is greater than (or equal to) c. Once again if it is true, then declare a as the largest. At this stage, if a is found to be smaller than c, then declare c as the largest. Here, notice that a has been already compared with b and has been found to be larger than b and hence c has to be larger than b.

If a was found to be smaller than b at the first comparison itself, then b has to be larger than a and now b has to be compared with c. As such, when the comparison of a with b gives a false (no) answer then b is compared with c to find out whether b is greater than (or equal to) c. If the answer here is true then b is declared as the biggest, otherwise (else) c is declared as the biggest.

Based on what's given, prepare flowchart and develop pseudo code.

Q7 - SUM OF FIRST N NATURAL NUMBERS

Here, the problem is to find the sum of the series $1 + 2 + 3 + 4 + 5 + 6 + 7 + \dots + n$.

The Method

This is one of the simplest examples one comes across to compute the sum of a series. It is simple because, the first term is 1 and each of the next terms are obtained by incrementing the previous term by 1 (i.e. by adding 1 to the previous term). A variable sum is used to hold the entire sum of the series. A variable like 'i' is used as a counter which counts from 1 to n in steps of 1. The value of this counter (i.e. variable) is added to sum at each step using a relation like $\text{sum} = \text{sum} + i$. The only input to this program is 'n', the value of number of terms.

Based on what's given, prepare flowchart and develop pseudo code.

Q8 - FINDING THE SUM OF ODD NUMBERS AND SUM OF EVEN NUMBERS FROM 1 TO N

The Method

Starting from 1, numbers 2, 3, 4 upto n are generated by adding (i.e. incrementing) a 1 to the previous number. As a sort of convention let the first number be term the initial value of which is 1. The next number (which is also called term) is generated by adding a 1 to the previous term (i.e. term). In other words, the relation $\text{term} = \text{term} + 1$ is used.

At every step, after generating a new term (i.e. term) it is tested to find out whether it is odd or even. For this modulo division by 2 is used (as discussed in 3.3). If the number is even then it is used to find the evensum and if the number is odd it is used to find the oddsum. Infact, evensum and oddsum are two variables, which are intially zero, and are specifically used for storing the required sum. The above procedure is repeated as long as the number (i.e. the term) is less than or equal to n. In other words, the above procedure is repeatedly done while the number (or term) is less than or equal to n.

Based on what's given, prepare flowchart and develop pseudo code.